

EXPERIENCE

Johns Hopkins University (JHU)

Postdoctoral fellow in the Department of Mechanical Engineering

– Focus: “Origin of vorticity in stratified flows.”

Baltimore, MD, USA

Feb. 2026 –Current

Georgia Institute of Technology (GT)

Postdoctoral fellow in Earth and Atmospheric Sciences

– Focus: “Interaction of ocean currents and seamounts”

Atlanta, GA, USA

Oct. 2023 –Jan. 2026

EDUCATION

University of New Hampshire (UNH)

Ph.D. in Applied Mathematics, GPA: 3.96/4.00, Advisor: Prof John F Gibson

– Thesis: “Symmetries, Bifurcations and Transition to Turbulence”

Durham, NH, USA

Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR)

M.S. in Engineering Mechanics, GPA: 6.90/8.00, Advisor: Prof M Alam

– Thesis: “Pattern Formation and Anomalous Modes in Axisymmetric Compressible Taylor-Couette Flow”

Bengaluru, India

Birla Institute of Technology and Science (BITS) Pilani

B.E.(Hons.) in Mechanical Engineering, GPA: 8.25/10.00

– Thesis: “Investigation of Turing Patterns Using Finite Element Method and Symmetry”

Pilani, India

PUBLICATIONS

- [1] **P. Aghor**, A. Bracco, and K. G. Sabra, “Influence of ocean model resolution on acoustic propagation near a seamount in the presence of internal waves”, *under review*, 2026.
- [2] **P. Aghor** and J. F. Gibson, “Symmetry based reduced order models for planar shear flows”, *under prep.*, 2026.
- [3] **P. Aghor**, M. McKinley, and A. Bracco, “Interaction of ocean currents and seamounts: Near-bottom dynamics around atlantis ii”, *Journal of Geophysical Research: Oceans*, vol. 131, no. 3, e2025JC023167, 2026.
- [4] **P. Aghor** and J. F. Gibson, “Invariant symmetric subspaces of plane Poiseuille flow”, *Journal of Fluid Mechanics*, 2025.
- [5] M. Atif, P. Dubey, **P. Aghor**, V. López-Marrero, T. Zhang, A. Sharfuddin, K. Yu, F. Yang, F. Ladeinde, Y. Liu, *et al.*, “Fourier neural operators for spatiotemporal dynamics in two-dimensional turbulence”, in *SC24-W: Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*, IEEE, 2024, pp. 41–48.
- [6] **P. Aghor** and M. Atif, “Effect of outer cylinder rotation on the radially heated Taylor-Couette flow”, *Physics of Fluids*, vol. 35, no. 9, 2023.
- [7] **P. Aghor** and M. Alam, “Nonlinear axisymmetric Taylor-Couette flow in a dilute gas: Multiroll transition and the role of compressibility”, *Journal of Fluid Mechanics*, vol. 909, 2021.

TEACHING

- **Teaching Assistant** at UNH 2018 –2019, 2021–2022
Linearity (covers ODE's, linear algebra, phase plane analysis), Multidimensional Calculus
- **Teaching Assistant** at BITS Pilani 2015
Finite Element Method (ME G512)

SCHOLARSHIPS AND AWARDS

- **Department of Mathematics and Statistics Teaching Assistant Award, UNH** 2022–2023
- **Dissertation Year Fellowship, UNH Graduate School Award** 2022–2023
- **Departmental Nominee, Graduate School TA Teaching Award** 2021–2022, 2022–2023
- **Research Assistant** 2019–2021
- **R. Narasimha Award for the Best MS Thesis in Engineering Mechanics** 2017–2018

CONFERENCES, WORKSHOPS, SUMMER SCHOOLS

Acoustical Society of America joint with International Congress on Acoustics May 2023
New Orleans, LA, USA

- Gave a talk on ‘Effect Of Bottom Topography On Sound Propagation Near The Atlantis II Seamount’

Rosbypalooza Jul 2024
University of Chicago

- Worked with Prof. William Boos (UC Berkley) on tropical stationary waves
- Worked with Prof. Da Yang and Prof. Dorian Abbot (UChicago) on a reduced model of extreme tropical precipitation

APS-DFD Meeting Nov 2023
Washington Convention Center Washington, DC, USA

- Gave a talk on ‘Instability Islands in the Radially Heated Taylor-Couette Flow’

Boulder Summer School - Hydrodynamics Across Scales Jul 2022
University of Colorado Boulder, Colorado, USA

- Poster presentation titled: ‘Invariant Subspaces of Channel Flow’

APS-DFD Meeting Nov 2021
Phoenix Convention Center Online

- Gave a talk on ‘Exploring Invariant Symmetry Subspaces of Channel Flow’

School on Dynamics of Complex Systems May-Jun 2016
International Center for Theoretical Sciences (ICTS, Bangalore) Bengaluru, Karnataka, India

- Theme - Geophysical Fluid Dynamics

CIMPA Summer School on Current Research in Finite Element Method Jun-Jul 2015
Indian Institute of Technology (IIT, Bombay) Mumbai, Maharashtra, India

- Conducted tutorial sessions on FreeFem++

Finite Element Meet 2014 Dec 2014
Tata Institute of Fundamental Research (TIFR-CAM) Bengaluru, Karnataka, India

- Gave a talk on ‘Numerical Continuation and Bifurcation in Presence of Symmetry in FreeFem++’

RELEVANT COURSEWORK

Fluid Mechanics, Asymptotics and Perturbation Methods, Physical Oceanography, Geophysical Fluid Dynamics, Spatiotemporal and Turbulent Dynamics, Algebra I (Group and Ring Theory), Waves in Fluids, Nonlinear Vibrations, Electrodynamics, Theory of Relativity, Statistical Mechanics, Mathematical Physics, High Performance Computing, Numerical Linear Algebra, Numerical PDE's, Chaosbook Part 1 and 2.

TEST SCORES

- **All India Rank: 7 out of 3292 candidates** in GATE 2015, Engineering Sciences

SKILLS

- **Languages:** Python, Julia, MATLAB, C++, C, FORTRAN
- **Open Source Solvers:** FreeFem++, Dedalus, Channelflow, AUTO -07p, Tensorflow
- **Miscellaneous:** OpenMP, MPI, high performance computing, scientific computing, machine learning, ocean models

REFERENCES:

1. Prof. John F. Gibson (john.gibson@unh.edu)
2. Prof. Gregory P. Chini (greg.chini@unh.edu)
3. Prof. Annalisa Bracco (annalisa.bracco@cmcc.it/annalisa@eas.gatech.edu)
4. Prof. Tamer A. Zaki (t.zaki@jhu.edu)
5. Teaching reference: Prof. Rita Hirschweiler (rita.hirschweiler@unh.edu)